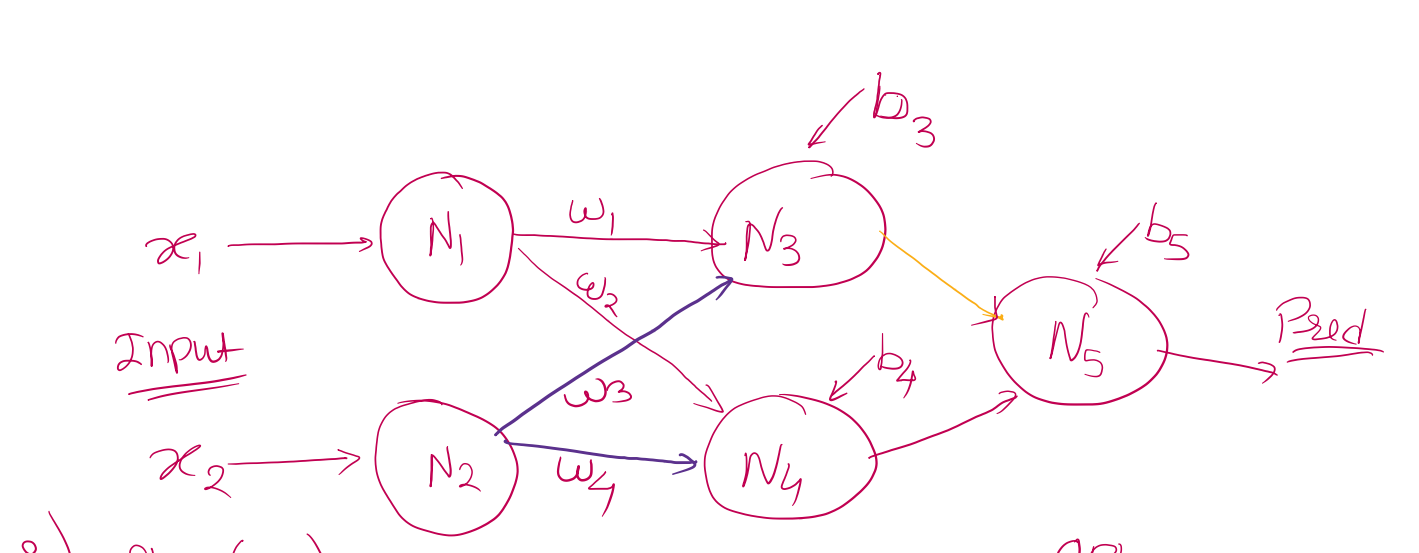
Forward Propogation



Study(hr) Sleep(hr)

x_1	x_2	Pass
4	6	1

$x_1 = 4$, $x_2 = 6$, $y = 1$

Initialize weights & biases

$w_1 \to 0.5$, $w_2 = 0.4$, $w_3 = 0.6$, $w_4 = 0.4$

$w_5 \to 0.6$ $\quad w_6 \to 0.3$

$b_3 \to 0.2$ $\quad b_4 \to 0.1$ $\quad b_5 = 0.3$

@ $\underline{N_3}$

$$Z_3 = w_1 x_1 + w_3 x_2 + b_3$$
$$= 0.5 \times 4 + 0.6 \times 6 + 0.2$$
$$= 2 + 3.6 + 0.2 = 5.8$$

$$A_3 = \frac{1}{1+e^{-Z_3}} = \frac{1}{1+e^{-5.8}} = 0.99698$$

@ $\underline{N_4}$

$$Z_4 = w_2 x_1 + w_4 x_2 + b_4$$
$$= 0.4 \times 4 + 0.4 \times 6 + 0.1$$
$$= 1.6 + 2.4 + 0.1$$
$$= 4.1$$

$$A_4 = \frac{1}{1+e^{-Z_4}} = \frac{1}{1+e^{-4.1}} = 0.98369$$

@ $\underline{N_5}$

$$Z_5 = w_5 \times A_3 + w_6 \times A_4 + b_5$$
$$= 0.6 \times 0.99698 + 0.3 \times 0.98369 + 0.3$$
$$= 1.193295$$

$$A_5 = \frac{1}{1+e^{-Z_5}} = \frac{1}{1+e^{-1.193295}} = \underline{\underline{0.7673}}$$

Backpropogation

Want to update $\underline{w_5}$?

$$\frac{dL}{dw_5} = \frac{\partial L}{\partial A_5} \times \frac{\partial A_5}{\partial Z_5} \times \frac{\partial Z_5}{\partial w_5} \text{ — \underline{Chain Rule}}$$

$$L = \text{log loss} = -\left[y\log(\hat{y}) + (1-y)\log(1-\hat{y}) \right]$$

$$\frac{\partial L}{\partial A} = \frac{\partial L}{\partial \hat{y}} = ?$$

* first term of loss function derivative *

$$= -y\log(\hat{y})$$
$$= -y/\hat{y}$$

* Derivative of Second term of Loss function *

$$= (1-y) \times \log(1-\hat{y})$$
$$= -(1-y) \times \frac{1}{(1-\hat{y})} \times \frac{d}{d\hat{y}}(1-\hat{y})$$
$$= -(1-y) \times \frac{1}{(1-\hat{y})} \times (0-1)$$
$$= -(1-y) \times \frac{1}{(1-\hat{y})} \times -1$$
$$= \frac{1-y}{1-\hat{y}}$$

$$\boxed{\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}}$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$\frac{\partial \hat{y}}{\partial z} = (1+e^{-z})^{-1}$$
$$= -1(1+e^{-z})^{-1-1} \frac{\partial}{\partial z}(1+e^{-z})$$
$$= -\frac{1}{(1+e^{-z})^2} \times \left(0 + e^{-z} \frac{\partial}{\partial z}(-z)\right)$$
$$= -\frac{1}{(1+e^{-z})^2} \times (e^{-z} \times -1)$$
$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

It can be simplied as

$$= \frac{1}{(1+e^{-z})} \times \frac{1}{(1+e^{-z})} \times e^{-z}$$
$$= \hat{y} \times (1-\hat{y})$$

$$1-\hat{y} = 1 - \frac{1}{1+e^{-z}}$$
$$= \frac{1+e^{-z}-1}{1+e^{-z}}$$
$$\Rightarrow \frac{e^{-z}}{1+e^{-z}}$$

* Now Come back to our question

$$\frac{\partial L}{\partial w_5} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial Z_5} \times \frac{\partial Z_5}{\partial w_5}$$

Two things are already Calculated lets calculate remaing $\frac{\partial Z_5}{\partial w_5}$.

$$Z_5 = w_5 \times A_3 + w_6 \times A_4 + b_5$$

$$\frac{\partial Z_5}{\partial w_5} = A_3 + 0 + 0$$

$$\frac{\partial Z_5}{\partial w_5} = A_3$$

$$\frac{\partial L}{\partial w_5} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial Z_5} \times \frac{\partial Z_5}{\partial w_5}$$
$$= \left(-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}\right) \times \hat{y}(1-\hat{y}) \times A_5$$
$$= \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\cancel{\hat{y}(1-\hat{y})}} \times \cancel{\hat{y}(1-\hat{y})} \times A_5$$
$$= (-y + \cancel{y\hat{y}} + \hat{y} - \cancel{y\hat{y}}) \times A_5$$

$$\boxed{\frac{\partial L}{\partial w_5} = (\hat{y} - y) \times A_5}$$

$$= (0.7673 - 1) \times 0.7673$$

$$\boxed{\frac{\partial L}{\partial w_5} = -0.1785}$$

$$w_{5_{new}} = w_{5_{old}} - \eta \times \frac{\partial L}{\partial w_5}$$
$$= 0.5 - 0.1 \times -0.1785$$
$$\boxed{w_{5\,new} = 0.51785}$$

* What if b_5 need update ?

$$\frac{\partial L}{\partial b_5} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial Z_5} \times \frac{\partial Z_5}{\partial b_5}$$

$$Z_5 = w_5 A_3 + w_6 A_4 + b_5$$
$$\frac{\partial Z_5}{\partial b_5} = 0 + 0 + 1$$
$$\underline{\underline{\frac{\partial Z_5}{\partial b_5} = 1}}$$

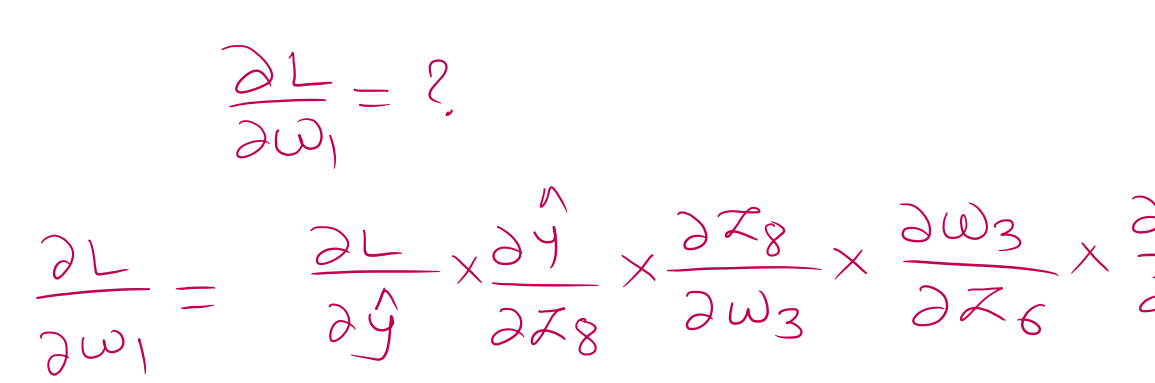
$$\frac{\partial L}{\partial b_5} = (\hat{y} - y) \times 1$$

$$\frac{\partial L}{\partial b_5} = 0.7673 - 1 = -0.2327$$

$$\boxed{\frac{\partial L}{\partial b_5} = -0.2327}$$

$$b_{5_{new}} = b_{5_{old}} - \eta \frac{\partial L}{\partial b_5}$$
$$= 0.3 - 0.1 \times -0.2327$$
$$\boxed{b_{5_{new}} = 0.32327}$$

* Some important points to Remember *



If need to update $w_1 = ?$

then w_1 contribute to loss in two way.

Path ① → $w_1 \to 3 \to w_2 \to 6 \to w_3 \to 8 \to \hat{y} \to$ Loss

Path ② → $w_1 \to 3 \to w_4 \to 7 \to w_5 \to 8 \to \hat{y} \to$ Loss

$$\frac{\partial L}{\partial w_1} = ?$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial Z_8} \times \frac{\partial Z_8}{\partial w_3} \times \frac{\partial w_3}{\partial Z_6} \times \frac{\partial Z_6}{\partial w_2} \times \frac{\partial w_2}{\partial Z_3} \times \frac{\partial Z_3}{\partial w_1}$$
$$+ \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial Z_8} \times \frac{\partial Z_8}{\partial w_5} \times \frac{\partial w_5}{\partial Z_7} \times \frac{\partial Z_7}{\partial w_4} \times \frac{\partial w_4}{\partial Z_3} \times \frac{\partial Z_3}{\partial w_1}$$

Here we Conclude our ANN forword Pass & Back propogation formally.

Thank You All.